COURSE CODE: **IFT 407**

COURSE TITLE: **PROFESSIONAL ETHICS, LEGAL AND SOCIAL ISSUES IN INFORMATION TECHNOLOGY**

COURSE UNITS: **(2 UNITS, COMPULSORY)**

LECTURER: **PROF. A. O. OGUNDE (ROOM 21, FCDT)**

COURSE CONTENT

Professional communications; teamwork concepts and issues; social context of computing – social informatics, social impact of information technology on society, online communities and social implications, philosophical context, diversity issues, gender related issues, cultural issues, accessibility issues, globalisation issues, economic issues in computing, and digital divide; intellectual property – foundations of intellectual property, ownership of information, plagiarism, software piracy, fair use, digital millennium copyright act, copyrights, patents, trademarks and trade secrets, international differences; legal issues in computing; organisational context; professional and ethical issues and responsibilities – relationships with professional societies, codes of professional conduct, ethics and history of ethics, whistle-blowing, workplace issues (harassment, discrimination), identify theft, ethical hacking; privacy and civil liberties.

SOME RECOMMENDED TEXTS

1. Contemporary Issues in Ethics and Information Technology by Robert A. Schultz, Woodbury University, USA. IRM Press.

2. Ethics in Information Technology by George W. Reynolds

PROFESSIONAL COMMUNICATIONS

Professional communication refers to the effective exchange of information, ideas, and meaning within a professional or organizational context.

In the field of Information Technology (IT), it is the bridge between technical expertise and ethical, legal, and social responsibility. An IT professional must be able not only to design and implement solutions but also to communicate clearly, responsibly, and ethically with clients, colleagues, and the public.

Meaning and Scope of Professional Communication

Professional communication encompasses all forms of verbal, non-verbal, and written interactions that occur in a workplace or professional environment.

It includes:

- Technical documentation
- Email and report writing
- Presentations and meetings
- Client communications
- Ethical disclosures and legal reporting

In the IT context, it also involves the communication of complex technical information in a way that is clear, accurate, and ethically appropriate to non-technical audiences.

Importance of Professional Communication in IT

1. **Ethical Responsibility:** Communicating truthfully and transparently supports ethical integrity and prevents misinformation or manipulation.

2. **Legal Compliance:** Many IT practices (such as data collection, user consent, and intellectual property use) require clear, legally compliant communication with stakeholders.
3. **Team Collaboration:** IT projects often involve cross-functional teams. Effective communication ensures coordination, reduces conflict, and promotes accountability.
4. **Public Trust:** Clear and honest communication helps build public confidence in digital systems, cybersecurity measures, and data governance.
5. **Crisis and Incident Reporting:** During security breaches or system failures, professional communication ensures timely, accurate, and lawful disclosure to relevant authorities.

Ethical Dimensions of Communication in IT

Professional communication must adhere to ethical standards, ensuring honesty, confidentiality, and respect for others. Key ethical principles include:

Ethical Principle	Application in IT Communication
Honesty	Avoiding false claims about systems, software capabilities, or data.
Confidentiality	Protecting sensitive data and proprietary information.
Transparency	Clearly stating system limitations, risks, or security implications.
Respect	Using inclusive and non-discriminatory language in professional discourse.
Accountability	Taking responsibility for communication errors or misrepresentations.

Legal Aspects of Professional Communication

In IT, poor or unethical communication can have legal consequences. Professionals must be aware of relevant laws and regulations guiding digital communication, such as:

- **Data Protection Laws (e.g., NDPR, GDPR):** Require clear communication of data usage, consent, and privacy rights.
- **Intellectual Property Laws:** Mandate proper acknowledgement of sources and authorship in documentation and coding.
- **Cybercrime Acts:** Prohibit false or malicious communication, phishing, and information manipulation.
- **Contractual Obligations:** Ensure that written agreements and emails constitute lawful, binding commitments.
- **Whistleblower Protection Laws:** Protect employees who communicate unethical or illegal practices in organisations.

Forms and Channels of Professional Communication in IT

Form	Description	Example
Written	Formal documents, reports, emails, policies	Project proposals, legal disclaimers
Oral	Verbal discussions, meetings, and interviews	Team briefings, client negotiations
Visual	Diagrams, charts, slides, data visualisation	System architecture diagrams
Digital	Electronic communication through ICT platforms	Slack, Teams, Zoom, corporate portals
Non-verbal	Tone, body language, and professional etiquette	Presentations, virtual meetings

Each mode must reflect professionalism, clarity, and ethical sensitivity.

Communication and Professional Codes of Conduct

Professional IT bodies such as IEEE, ACM, CPN and NCS emphasise communication ethics in their codes of conduct. Key provisions include:

- Communicating technical information truthfully and objectively.
- Respecting confidentiality and intellectual property.
- Reporting unethical or illegal activities responsibly.
- Ensuring fairness and non-discrimination in all professional interactions.

Challenges in Professional Communication

Common barriers include:

- Technical jargon confusing non-technical audiences.
- Cultural and language differences in global teams.
- Information overload from digital communication channels.
- Ethical dilemmas when organisational interests conflict with public safety.
- Misinterpretation of tone in electronic communication.

Overcoming these challenges requires emotional intelligence, ethical awareness, and adherence to professional standards.

Best Practices for Ethical and Legal Professional Communication

1. Communicate truthfully and accurately.
2. Maintain confidentiality of sensitive data.
3. Use professional, inclusive, and respectful language.
4. Cite sources and respect intellectual property.
5. Obtain and document consent for data use.

6. Keep records of official communications.
7. Follow organisational and legal reporting protocols.
8. Avoid plagiarism and misinformation.
9. Provide timely, clear, and complete information in all reports.
10. Practice digital etiquette and cyber hygiene in communication tools.

In summary, professional communication is an ethical, legal, and social responsibility of every IT practitioner. Beyond transmitting information, it reflects one's integrity, accountability, and commitment to professional excellence. An IT professional who communicates clearly, ethically, and lawfully not only enhances organisational credibility but also promotes trust, security, and good governance in the digital society.

TEAMWORK CONCEPTS AND ISSUES

Teamwork is a foundational element of professional practice in the field of Information Technology (IT). Modern IT projects—such as system design, software engineering, cybersecurity operations, and data analytics—are often multidisciplinary and require close collaboration among professionals.

In the context of professional ethics and legal issues, teamwork is not only a matter of efficiency but also of ethical responsibility, accountability, and compliance with professional standards.

Effective teamwork ensures that:

- Ethical principles guide collaborative decision-making.
- Legal obligations are respected in group projects.
- Technical outcomes reflect professionalism and shared integrity.

In the rapidly evolving world of Information Technology, teamwork stands as a pillar of ethical professionalism and lawful practice. While technical expertise drives innovation, collaboration rooted in ethics, transparency, and legal responsibility ensures sustainable and trustworthy digital solutions.

Every IT professional must, therefore, cultivate the ability to work ethically in teams—upholding integrity, respecting the law, and contributing to a culture of excellence and accountability.

Meaning of Teamwork

Teamwork refers to the coordinated and cooperative efforts of individuals working together towards a common goal. It involves the integration of diverse skills, ideas, and experiences to solve problems and achieve results efficiently and ethically.

In IT, teamwork may take place within:

- Software development teams

- IT support and network management units
- Cybersecurity response teams
- Academic research collaborations
- Project management and system implementation groups

Characteristics of an Effective IT Team

An effective IT team displays the following characteristics:

1. **Clear Objectives:** Shared understanding of project goals and success criteria.
2. **Complementary Skills:** Members possess technical, analytical, ethical, and communication competencies.
3. **Mutual Respect:** Recognition of each member's contribution regardless of hierarchy or background.
4. **Effective Communication:** Open, respectful, and timely sharing of information and feedback.
5. **Ethical Commitment:** Adherence to professional codes of conduct and confidentiality standards.
6. **Accountability:** Shared responsibility for team outcomes—successes or failures.
7. **Conflict Resolution Mechanism:** Ethical and constructive approaches to addressing disagreements.

The Ethical Dimensions of Teamwork in IT

Ethics plays a vital role in ensuring fairness, honesty, and respect within teams.

Key ethical principles guiding teamwork include:

Ethical Principle	Description	Example in IT Context
Integrity	Acting honestly and truthfully in team contributions	Accurately reporting coding progress or test results
Confidentiality	Respecting sensitive data and internal project information	Not disclosing client data to outsiders
Fairness	Giving credit and responsibility where due	Recognizing every contributor in documentation
Accountability	Owning up to errors and correcting them responsibly	Reporting a security flaw rather than concealing it
Respect	Valuing others' opinions and diversity	Avoiding discrimination or harassment in teams

A team that upholds these ethical principles fosters trust, collaboration, and credibility.

Legal Issues Related to Teamwork in IT

While teamwork is primarily collaborative, it also carries legal implications, particularly in professional IT practice. Common legal considerations include:

1. **Intellectual Property (IP) Rights:** Determining ownership of jointly developed software, algorithms, or research outputs.
2. **Confidentiality Agreements (NDAs):** Legal documents that protect sensitive team or client information from unauthorized disclosure.
3. **Liability and Responsibility:** Defining who is legally accountable if a system fails or violates user rights.
4. **Data Protection and Privacy Laws:** Ensuring that teams handling user data comply with laws like the NDPR, GDPR, or Cybercrime Act.

5. **Workplace Harassment and Discrimination Laws:** Safeguarding ethical and inclusive teamwork environments, free from bullying, bias, or gender discrimination.
6. **Contractual Obligations:** Clarifying team roles, timelines, and deliverables within legal agreements.

Common Issues and Challenges in IT Teamwork

Issue	**Description**	**Ethical/Legal Implication**
Poor Communication	Misunderstandings or lack of updates	Leads to project delays, errors, or ethical breaches
Unequal Participation	Some members dominate while others are passive	Raises fairness and accountability concerns
Conflict of Interest	Personal or financial interests influencing decisions	May violate professional ethics and organizational policy
Intellectual Property Disputes	Ownership of ideas, software, or publications	Can lead to legal action or loss of trust
Cultural and Personality Differences	Diversity-related friction or bias	Must be managed ethically to ensure inclusivity
Unethical Collaboration	Plagiarism, falsification, or code theft	Breach of both ethical codes and legal standards

Models of Effective Teamwork in IT

Several frameworks promote teamwork excellence and ethical collaboration:

1. **Agile and Scrum Methodologies:**
   Encourage continuous communication, transparency, and shared accountability.

2. **Team Chartering:**
   Creating a formal document that defines team values, rules, and responsibilities.
3. **Peer Review and Code Review:**
   Promotes accuracy, accountability, and integrity in technical outputs.
4. **Diversity and Inclusion Programs:**
   Encourage participation from all backgrounds and ensure fair treatment.
5. **Ethical Leadership:**
   Team leaders model honesty, fairness, and respect, setting the moral tone for others.

Strategies for Managing Teamwork Issues

To ensure ethical and legally compliant teamwork, IT professionals should adopt the following strategies:

1. Set clear ethical guidelines at project initiation.
2. Encourage transparent communication among all team members.
3. Establish reporting and feedback systems to address issues early.
4. Document all team agreements, contributions, and versions for legal accountability.
5. Train members on ethical conduct, data protection, and IP rights.
6. Foster inclusion and mutual respect to promote psychological safety.
7. Address conflicts constructively, focusing on solutions rather than blame.

Relevance of Teamwork to Professional Ethics and Legal Compliance

Teamwork is central to the ethical and legal framework of IT practice. Through teamwork:

- Professionals learn to balance individual responsibility with collective accountability.
- Organizations ensure compliance with codes of ethics (e.g., IEEE, ACM, CPN, NCS).
- Teams enhance quality assurance and risk management, reducing chances of ethical or legal violations.

Ethical teamwork, therefore, becomes a professional obligation, not merely a managerial preference.

SOCIAL CONTEXT OF COMPUTING

The social context of computing explores how computing technologies interact with human values, ethics, culture, and societal structures. Information Technology (IT) does not exist in isolation — it influences society and is, in turn, shaped by social, philosophical, and cultural factors. For IT professionals, understanding this context is critical to making ethical and lawful decisions that promote fairness, inclusivity, and responsible innovation. The social context of computing highlights that technology is never neutral — it carries ethical, cultural, economic, and legal consequences.

Professionals in the field of IT must, therefore:

- Act with social responsibility.
- Uphold ethical standards in the design, use, and governance of technology.
- Strive to ensure that computing serves humanity equitably, bridging divides rather than deepening them.

In essence, responsible computing is not just about code and systems, but about people, values, and justice in the digital world.

Social Informatics

Social informatics is the study of the social aspects of computerisation, including how people use and are affected by information and communication technologies (ICTs).

Key Concepts

- Focuses on the interaction between technology, people, and institutions.
- Examines social, organisational, and policy implications of information systems.

- Emphasises that technological systems are socially constructed and context-dependent.

Relevance to Professional Ethics

- Encourages IT professionals to design systems that are user-centred and socially responsible.
- Promotes ethical reflection on how IT affects work practices, privacy, and digital inclusion.

Social Impact of Information Technology on Society

The widespread adoption of IT has transformed virtually every aspect of human life — communication, education, commerce, governance, and entertainment.

Positive Impacts

- Enhanced access to information and education.
- Improved efficiency in industries and governance.
- Empowerment through digital entrepreneurship.

Negative Impacts

- Cybercrime and data breaches.
- Privacy invasion and surveillance.
- Job displacement through automation.
- Social isolation and misinformation.

Ethical Implications

IT professionals must balance innovation with responsibility, ensuring that technologies benefit humanity without causing harm or inequality.

Online Communities and Social Implications

Online communities are virtual groups formed around shared interests, goals, or identities, often through platforms such as social media, forums, or professional networks.

Social and Ethical Implications

- Facilitate collaboration and knowledge sharing, but also spread misinformation and cyberbullying.
- Raise issues of digital identity, privacy, and freedom of expression.
- Create echo chambers that influence political and social opinions.
- May require moderation policies to ensure ethical interaction and legal compliance (e.g., data protection, hate speech laws).

IT professionals have a duty to design and manage online systems responsibly, ensuring safety, fairness, and inclusivity.

Philosophical Context of Computing

The philosophy of computing deals with questions about the nature, purpose, and consequences of computation and artificial intelligence.

Philosophical Questions

- What does it mean for a machine to "think"?
- Should AI systems have moral or legal status?
- How do we define ethical responsibility in autonomous systems?

Ethical Relevance

Philosophical inquiry helps IT professionals to:

- Examine the moral implications of emerging technologies.

- Distinguish between technological determinism (technology controls society) and social constructivism (society shapes technology).
- Develop value-sensitive designs that integrate human welfare and rights.

Diversity Issues in Computing

Diversity in computing refers to the inclusion of people from different backgrounds, races, genders, abilities, and perspectives in the IT field.

Challenges

- Underrepresentation of certain groups (e.g., minorities, persons with disabilities).
- Biased algorithms resulting from non-diverse design teams.
- Workplace discrimination and exclusion in tech environments.

Ethical and Legal Considerations

- Promoting equal opportunity in hiring and project participation.
- Enforcing anti-discrimination laws in tech workplaces.
- Designing technologies that serve all user demographics fairly.

Gender-Related Issues In Computing

Gender issues persist in computing due to historical biases and unequal representation.

Key Issues

- Gender stereotypes that discourage women from ICT careers.
- Gender pay gaps and limited leadership roles for women in tech.
- Online harassment and gender-based digital violence.

Ethical and Legal Responses

- Promoting gender equity in education and workplace policies.
- Enforcing laws on gender discrimination and online abuse.
- Encouraging inclusive system design that avoids gender bias in AI or software outputs.

Cultural Issues In Computing

Computing systems reflect the values, language, and norms of their creators.
When deployed globally, such systems can clash with cultural expectations or reinforce biases.

Examples

- Software localisation and multilingual support.
- Respecting cultural norms in digital content and advertisements.
- Avoiding ethnocentric assumptions in design (e.g., date formats, icons, colour meanings).

Ethical Principle

Culturally sensitive computing ensures that technology respects human diversity, supports cross-cultural communication, and avoids digital imperialism.

Accessibility Issues in Computing

Accessibility means designing IT systems and digital environments usable by everyone, including persons with disabilities.

Accessibility Challenges

- Inaccessible websites or software interfaces.

- Lack of assistive technologies (e.g., screen readers, captioning tools).
- Neglect of inclusive design principles.

Legal and Ethical Frameworks

- UN Convention on the Rights of Persons with Disabilities (CRPD).
- Web Content Accessibility Guidelines (WCAG).
- Ethical responsibility to ensure equal access to digital resources.

Globalisation Issues In Computing

Globalisation has transformed IT into a borderless ecosystem of communication, commerce, and innovation.

Opportunities

- Global collaboration and open-source development.
- Access to international markets and remote work opportunities.

Challenges

- Cross-border data privacy and jurisdictional conflicts.
- Outsourcing and labour exploitation.
- Cultural homogenization and loss of local identity.

Ethical Imperative

IT professionals must operate within international ethical standards, respecting global digital rights, intellectual property laws, and fair trade practices.

Economic Issues In Computing

Computing technologies are both economic drivers and disruptors.

Key Economic Issues

- Automation leading to job loss and skill displacement.
- Intellectual property and software licensing costs.
- Unequal access to digital infrastructure in developing economies.
- The rise of monopolistic digital corporations.

Ethical and Legal Relevance

- Need for fair competition, open innovation, and data ownership rights.
- Corporate responsibility in promoting economic inclusivity through technology.

Digital Divide

The digital divide refers to the gap between individuals and societies with access to modern ICTs and those without.

Dimensions

- **Access Divide:** Lack of infrastructure or devices.
- **Skill Divide:** Inadequate digital literacy.
- **Usage Divide:** Limited or inequitable use of technology.

Ethical and Social Responsibility

IT professionals and policymakers should:

- Promote universal access to ICTs.
- Support digital literacy education.
- Develop affordable and inclusive technologies.

Bridging the digital divide is both an ethical duty and a social investment for global equity.

INTELLECTUAL PROPERTY – FOUNDATIONS OF INTELLECTUAL PROPERTY

Intellectual property is a term used to describe works of the mind—such as art, books, films, formulas, inventions, music, and processes—that are distinct and owned or created by a single person or group.

Intellectual property is protected through copyright, patent, and trade secret laws.

- **Copyright law** protects authored works, such as art, books, film, and music.
- **Patent law** protects inventions.
- **Trade secret law** helps safeguard information that is critical to an organisation's success.

COPYRIGHTS

A **copyright** grants the creators of original works the exclusive right to distribute, display, perform, or reproduce the work in copies, or to prepare a derivative work.

Copyright infringement is a violation of the rights secured by the owner of a copyright, which occurs when someone copies a substantial and material part of another's copyrighted work without permission.

Copyright Term

- Copyright law guarantees developers the rights to their works for a certain amount of time. Since 1960, the term of copyright has been extended 11 times from its original limit of 28 years.
- The current copyright term is life of the author plus 70 years.

Eligible Works for Copyright

- The types of work that can be copyrighted include architecture, art, audiovisual works, choreography, drama, graphics, literature, motion pictures, music, pantomimes, pictures, sculptures, sound recordings, and other intellectual works.
- To be eligible for copyright, a work must fall within one of the identified categories, and it must be original.
- An idea cannot be copyrighted, but the expression of an idea can be copyrighted.

Fair Use Doctrine

The fair use doctrine allows portions of copyrighted materials to be used without permission under certain circumstances.

Courts consider the following four factors when deciding whether a particular use of copyrighted property is fair and can be allowed without penalty:

1. The purpose and character of the use (non-commercial versus commercial use).
2. The nature of the copyrighted work.
3. The portion of the copyrighted work used in relation to the work as a whole.
4. The effect of the use on the value of the copyrighted work.

Software Copyright Protection

The use of copyrights to protect computer software raises many complicated issues of interpretation with regard to allegations of copyright infringement. For example, to prove infringement, software copyright holders must show a striking resemblance between their software and another developer's new software that could be explained only by copying. This is often difficult because two developers can conceivably develop separate but nearly identical programs to perform simple tasks.

The Prioritizing Resources and Organization for Intellectual Property Act

1. The Prioritizing Resources and Organization for Intellectual Property (PROIP) Act of 2008 increased trademark and copyright enforcement and substantially increased penalties for infringement.

2. The law also established the Office of the United States Intellectual Property Enforcement Representative within the U.S. Department of Justice.

General Agreement on Tariffs and Trade (GATT)

1. The GATT is a trade agreement between 117 countries, and it created the World Trade Organisation (WTO) to enforce its policies.

2. GATT includes a section covering copyrights called the Agreement on Trade-Related Aspects of Intellectual Property Rights (TRIPS).

The World Intellectual Property Organisation (WIPO) Copyright Treaty

1. The WIPO Copyright Treaty, adopted in 1996, provides additional copyright protections to address electronic media.

2. The treaty ensures that computer programs are protected as literary works and that the arrangement and selection of material in databases is also protected.

The Digital Millennium Copyright Act (1998)

The Digital Millennium Copyright Act (DMCA) was signed into law in 1998 and implements two 1996 WIPO treaties: the WIPO Copyright Treaty and the WIPO Performances and Phonograms Treaty.

The act is divided into five sections, Titles I–V. Title II provides “safe harbors” for ISPs whose customers/subscribers might be breaking

copyright laws by downloading, posting, storing, or sending copyrighted material via its services.

There are two competing views about the DMCA. Some people view the DMCA as a boon to the growth of the Internet and its use as a conduit for innovation and freedom of expression. Without the safe harbours that the DMCA provides, the risk of copyright liability would be so great as to seriously discourage ISPs from hosting and transmitting user-generated content. Other people view the DMCA as extending too much power to copyright holders that it restricts the flow of information.

PATENTS

A **patent** is a grant of a property right issued by the United States Patent and Trademark Office (USPTO) to an inventor. A patent permits its owner to exclude the public from making, using, or selling a protected invention, and it allows for legal action against violators. Unlike a copyright, a patent prevents independent creation as well as copying. To be patentable, an invention has to be new (not found in prior art), useful, and not obvious to a **p**erson **h**aving **o**rdinary **s**kill **i**n **t**he **a**rt (PHOSITA). In addition, the patent application must include a written description of the invention, and the description must enable a PHOSITA to make and use the invention without undue experimentation.

An invention is new only if it cannot be found in prior art. Prior art is the existing body of knowledge that is available to a person of ordinary skill in the art.

Abstract ideas, laws of nature, and natural phenomena cannot be patented.

Patent infringement is a violation of the rights secured by the owner of a patent, which occurs when someone makes unauthorised use of another's patent. Patent infringement can be direct or indirect infringement.

Software Patents

Prior to 1981, courts regularly turned down requests to patent software, giving the impression that software could only be copyrighted but not patented. However, in the 1981 Diamond V. Diehr case, the Supreme Court granted a patent to Diehr, who had developed a process control computer and sensors to monitor the temperature inside a rubber mould. Based on this ruling, the courts have slowly broadened the scope of patent protection for software-related inventions.

Some experts believe that too many software patents are being granted, and that this inhibits new software development, as evidenced by the number of software patent infringement lawsuits between large software organisations such as Oracle, Google, and Apple.

Cross-Licensing Agreements

Many large software companies have cross-licensing agreements in which each party agrees not to sue the other over patent infringements. Major IT firms usually have little interest in cross-licensing with smaller firms, so small businesses often have no choice but to license patents if they use them.

TRADE SECRETS

A **trade secret** is defined as business information that represents something of economic value, has required effort or cost to develop, has some degree of uniqueness or novelty, is generally unknown to the public, and is kept confidential.

Trade secrets have the following key advantages.

Trade Secret Laws

Trade secret protection laws vary greatly from country to country. For example, there is no legal protection for trade secrets in the

Philippines. It is, therefore, important for information managers to understand the legal protections available, especially when doing business in foreign countries.

Employees and Trade Secrets

Employees pose the greatest threat to the loss of company trade secrets—they may accidentally disclose or steal them for personal gain.

Organisations must educate employees about the importance of maintaining the secrecy of corporate information.

Organisations often try to prohibit departing employees from revealing secrets by adding nondisclosure clauses to employment contracts. A typical nondisclosure clause for software companies is a clause in an employment contract that states that an employee cannot take copies of computer programs or reveal the details of software owned by the firm, even when they leave.

Employers can also use noncompete agreements to protect intellectual property from being used by competitors when key employees leave. A noncompete agreement prohibits an employee from working for any competitor for a period of time, often one to two years.

Judicial treatment of noncompetent agreements varies from state to state. Thus, Information managers should be aware of how courts in their jurisdiction treat noncompete agreements.

KEY INTELLECTUAL PROPERTY ISSUES

The next section discusses legal and ethical issues related to intellectual property and information technology, including plagiarism, reverse engineering, open-source code, competitive intelligence, trademark infringement, and cybersquatting.

A. Plagiarism

- Plagiarism is the act of stealing someone's ideas or words and passing them off as one's own.
- The explosion of electronic content and the growth of the Internet have made it easy to cut and paste paragraphs into term papers and other documents without proper citation or quotation marks.
- However, plagiarism detection services and software allow teachers, corporations, law firms, and publishers to check for matching text in different documents as a means of identifying potential plagiarism.

B. Reverse Engineering

Reverse engineering is the process of taking something apart in order to understand it, build a copy of it, or improve it. Reverse engineering was originally applied to computer hardware but is now commonly applied to software as well. A frequent use of reverse engineering for software is to modify an application that ran on one vendor's database so that it can run on another's (for example, from Access to Oracle).

Software license agreements increasingly forbid reverse engineering. As a result of the increased legislation affecting reverse engineering, some software developers are moving their reverse-engineering projects offshore to avoid U.S. rules. Reverse engineering introduces several ethical issues for consideration.

a. Some argue that its use is fair if it enables a company to create software that interoperates with another company's software or hardware and provides a useful function.

b. Others argue that it can uncover software designs that someone else developed at great research and development cost, thereby robbing the creator of future earnings and consequently reducing the business incentive for software development.

C. Open-Source Code

Open source code is any program whose source code is made available for use or modification, as users or other developers see fit. The basic premise behind open source code is that when many programmers can read, redistribute, and modify a program's code, the software improves.

Examples of open-source software include the following.

- Operating Systems e.g. Android, Ubuntu, Linux
- Internet Browsers e.g. Mozilla Firefox, Chromium
- Development Tools e.g. Vs code (Visual Studio Code), Android Studio, PyCharm, Xcode
- Productivity Software e.g. LibreOffice, OpenOffice
- Graphics and Design e.g. GIMP, Inkscape, Blender
- Web Development Frameworks e.g. Django, PyTorch (Python package for neural nets and machine learning)
- Collaboration Tools e.g. Nextcloud, Mattermost
- Content Management Systems e.g. Strapi, WordPress

D. Competitive Intelligence

Competitive intelligence is legally obtained information that is gathered to help a company gain an advantage over its rivals. An effective competitive intelligence operation requires the continual gathering, analysis, and evaluation of data with controlled dissemination of useful information to decision makers. Competitive intelligence is often integrated into a company's strategic plan and decision making.

Industrial espionage is the gathering of information not available to the public through illegal means. Industrial espionage is a serious crime that carries heavy penalties.

Competitive intelligence analysts must avoid unethical or illegal actions, such as lying, misrepresentation, theft, bribery, or eavesdropping with illegal devices when gathering data for competitive intelligence.

E. Trademark Infringement

- A trademark is a logo, package design, phrase, sound, or word that enables a consumer to differentiate one company's products from another's.
- Consumers often cannot examine goods or services to determine their quality or source, so instead they rely on the labels attached to the products. Thus, a trademark is anything that enables a consumer to differentiate one company's products from another's.
- A trademark holder has the right to prevent others from using the same mark or a confusingly similar mark on a product's label.
- Trademark infringement lawsuits often result from alleged uses of plaintiff's trademark in a website or a domain name.
- However, nominative fair use is a defence often employed by the defendant in trademark infringement cases where the defendant uses a plaintiff's mark to identify the plaintiff's products or services in conjunction with its own product or services.

F. Cybersquatting

A **cybersquatter** is a person or company that registers domain names for famous trademarks or company names to which they have no connection, with the hope that the trademark's owner will buy the domain name for a large sum of money.

The Anticybersquatting Consumer Protection Act (ACPA), enacted in 1999, allows trademark owners to challenge foreign cybersquatters who might otherwise be beyond the jurisdiction of U.S. courts.

The ACPA also allows trademark owners to challenge the registration of their trademark as a domain name even if the trademark owner has not created an actual website.

INTERNATIONAL DIFFERENCES

In today's globalized digital environment, computing and information technology (IT) practices transcend national borders. However, ethical standards, cultural values, and legal systems differ significantly across nations. These international differences influence how professionals, organizations, and governments interpret and enforce laws, regulations, and ethical responsibilities in the use and management of information technologies. Understanding these differences is essential for IT professionals who work in multinational environments or handle global data transactions.

International differences in computing refer to the variations in ethical norms, cultural expectations, and legal frameworks that govern information technology across countries. These differences affect:

- How privacy is defined and protected.
- How intellectual property is treated.
- What constitutes acceptable online behavior.
- How cybersecurity laws are enforced.
- How data can be stored, processed, or transferred internationally.

Thus, IT professionals must be aware of local and global legal variations when developing, deploying, or managing technological systems.

Areas of International Differences in IT Ethics and Law

a. Data Protection and Privacy Laws

Different countries have distinct **data protection laws** governing how personal data is collected, stored, and shared.

- **European Union (EU):** The General Data Protection Regulation (GDPR) is one of the strictest data privacy laws, emphasising user consent, data minimisation, and transparency.

- **United States (US):** Data privacy laws are more sectoral, varying by industry (e.g., HIPAA for healthcare, COPPA for children's data).
- **Nigeria:** The **Nigeria Data Protection Regulation (NDPR)** and the **Nigeria Data Protection Act (NDPA 2023)** govern local data protection but differ in enforcement capacity.
- **Asia:** Countries like China and India have emerging but highly state-regulated frameworks, often prioritising national security over individual privacy.

Ethical Implication:

IT professionals must ensure compliance with applicable jurisdictional laws when handling cross-border data transfers.

b. Cybercrime and Computer Misuse Laws

Cybercrime definitions and punishments vary internationally.

- The **Budapest Convention on Cybercrime (2001)** is the first international treaty aimed at harmonising cybercrime laws across nations.
- Countries like Nigeria enforce local legislation such as the **Cybercrimes (Prohibition, Prevention, etc.) Act, 2015**.
- In the United States, multiple laws, such as the Computer Fraud and Abuse Act (CFAA) criminalise unauthorised access.
- Some countries have broad definitions of cybercrime that may include political speech, raising ethical concerns about **freedom of expression**.

Ethical Implication: What is legal in one jurisdiction (e.g., encryption use, anonymous speech) might be illegal in another, requiring ethical sensitivity and legal awareness.

c. Intellectual Property Rights (IPR)

Software development and content creation often cross borders, leading to complex intellectual property disputes.

- Copyright, patent, and trademark laws differ by country.
- The World Intellectual Property Organization (WIPO) promotes global standards but enforcement varies.
- Issues like software piracy, open-source licensing, and digital content ownership are often treated differently in developing vs. developed nations.

Ethical Implication: IT professionals must respect IP rights across jurisdictions and avoid unauthorized replication or distribution of digital products.

d. Freedom of Information and Censorship

- Western democracies often promote freedom of information and open internet access.
- Authoritarian regimes may impose internet censorship, content filtering, or surveillance to control digital communication.
- Social media regulations vary widely — e.g., the EU Digital Services Act (2024) vs. China's Cybersecurity Law (2017).

Ethical Implication: IT professionals must balance compliance with national laws and the ethical duty to protect freedom of expression and human rights.

e. Cultural and Ethical Variations

Cultural beliefs influence how ethical principles are interpreted in computing.

- **Individualistic societies** (e.g., USA, UK) emphasize personal privacy and autonomy.
- **Collectivist societies** (e.g., China, Japan) may prioritize community harmony and state security.

- Perceptions of acceptable online behavior, data sharing, and intellectual property differ based on cultural norms.

Ethical Implication: IT professionals must demonstrate **cultural competence**, recognizing that ethical behavior is context-dependent while maintaining universal moral standards such as honesty, fairness, and respect.

f. Accessibility and Inclusion Standards

Accessibility requirements for digital products differ globally.

- **Web Content Accessibility Guidelines (WCAG)** provide international standards, but adoption varies.
- Some countries enforce accessibility laws (e.g., **Americans with Disabilities Act (ADA)** in the US), while others rely on voluntary compliance.

Ethical Implication: Developers must ensure that technology products are inclusive and accessible, regardless of regional laws, upholding global ethical responsibility.

g. Cross-Border Data Flow and Cloud Computing

- Cloud computing raises concerns over data sovereignty — who owns and controls data stored on foreign servers.
- Some nations restrict data transfer to foreign servers (e.g., data localization policies in India and Russia).
- International agreements such as the EU–US Data Privacy Framework (2023) aim to regulate transatlantic data transfers.

Ethical Implication: IT practitioners must ensure transparency, consent, and lawful transfer of data across borders.

Challenges Arising from International Differences

1. **Conflict of Laws:** Different jurisdictions may impose contradictory requirements.

2. **Jurisdictional Ambiguity:** Difficulty determining which nation's law applies to an online crime or contract.
3. **Varying Enforcement Strength:** Some countries lack the infrastructure to enforce cyber laws effectively.
4. **Cultural Misunderstandings:** Differing norms about privacy, freedom, and communication may cause ethical conflicts.
5. **Digital Colonialism:** Dominance of technology by global corporations can lead to economic and ethical inequalities.

Professional and Ethical Responsibilities

An IT professional working across borders should:

1. Understand **international legal frameworks** relevant to their work.
2. Comply with both **local** and **global** regulations where applicable.
3. Uphold **universal ethical standards** even where laws are silent or ambiguous.
4. Promote **fairness, inclusivity, and cultural sensitivity** in digital environments.
5. Advocate for **international cooperation** on cybersecurity, data protection, and human rights in computing.

In summary, international differences in ethics and law significantly shape the practice of computing in the global digital economy. While laws may differ from one country to another, core ethical values—honesty, respect, fairness, accountability, and responsibility—remain universal. A globally competent IT professional must therefore navigate these differences with ethical sensitivity, cultural awareness, and a commitment to professional integrity, ensuring that technology serves humanity responsibly across borders.

LEGAL ISSUES IN COMPUTING

The rise of the Information Age has introduced tremendous opportunities for innovation, but also complex legal challenges. As computing technologies permeate all aspects of human life—commerce, education, healthcare, government, and entertainment—legal frameworks are needed to regulate how information is created, stored, transmitted, and used. Legal issues in computing refer to the laws, regulations, and ethical obligations that govern the use, development, and management of information technology systems. For IT professionals, understanding these laws is essential for ensuring compliance, accountability, and responsible practice.

Legal issues in computing are central to professional ethics and responsible IT practice. Every computing professional must understand the legal environment governing digital operations, respect intellectual property, protect privacy, and prevent misuse of technology. Ethical and legal compliance not only prevent violations but also build public trust, enhance professional reputation, and promote sustainable digital development. Ultimately, law and ethics together form the backbone of professional integrity in the information technology industry.

Importance of Legal Awareness in IT Practice

Legal literacy in computing ensures that professionals:

- Operate within the boundaries of law and professional ethics.
- Prevent litigation and reputational damage due to unlawful actions.
- Protect intellectual property, data privacy, and users' rights.
- Promote safe and secure technology use in organisations.

An IT professional who ignores legal responsibilities risks criminal liability, civil penalties, and ethical violations.

Major Legal Areas in Computing

Legal issues in computing can be broadly categorised into several interrelated domains:

A. Data Protection and Privacy Laws

Data privacy laws regulate how personal and sensitive information is collected, processed, stored, and shared.

Key Concepts

- **Personal Data:** Any information relating to an identified or identifiable person.
- **Data Controller & Processor:** Entities responsible for managing or handling user data.
- **User Consent:** Legal requirement before data is collected or used.

Relevant Legal Frameworks

- **Nigeria Data Protection Regulation (NDPR)** – Protects the rights of Nigerian citizens' data.
- **General Data Protection Regulation (GDPR)** – European Union law influencing global data protection practices.
- **Cybercrime (Prohibition, Prevention, etc.) Act (2015)** – Addresses unauthorized access and data misuse.

Ethical Dimension

IT professionals must handle data lawfully, fairly, and transparently, ensuring confidentiality and accountability.

B. Intellectual Property (IP) Rights

Intellectual property protects creative works and innovations in computing—such as software, databases, and algorithms.

Types of IP Rights

- **Copyright:** Protects software codes, documentation, and digital content.

- **Patents:** Cover novel technological inventions and processes.
- **Trademarks:** Protect brand identity (names, symbols, logos).
- **Trade Secrets:** Safeguard confidential business information and algorithms.

Legal and Ethical Relevance

- Unauthorised software copying, code plagiarism, or reverse engineering violates IP laws.
- IT professionals must respect ownership and licensing agreements to avoid legal disputes.

C. Cybercrime and Computer Misuse

Cybercrime refers to criminal activities carried out using or targeting computers, networks, or digital systems.

Examples of Cybercrime

- Hacking and unauthorised access.
- Identity theft and phishing.
- Malware creation and distribution.
- Online fraud, scams, and financial crimes.
- Denial of Service (DoS) attacks.

Applicable Laws

- Cybercrime (Prohibition, Prevention, etc.) Act, 2015 (Nigeria).
- Computer Misuse Act (UK).
- Budapest Convention on Cybercrime (International).

Ethical Responsibility

IT professionals must uphold system security, avoid misuse of access privileges, and report cyber threats responsibly.

D. Computer Contracts and Software Licensing

Contracts govern relationships between IT professionals, clients, and vendors.

Legal Issues

- **Software Licensing Agreements:** Define how users can install, modify, or redistribute software.
- **Service Level Agreements (SLAs):** Specify the quality and scope of IT services.
- **Breach of Contract:** Occurs when terms are violated, leading to legal consequences.

Best Practices

- Read and understand licensing terms before software deployment.
- Maintain proper documentation for contracts and compliance audits.
- Avoid using pirated or unlicensed software.

E. Cybersecurity and Legal Accountability

Cybersecurity laws define how individuals and organizations must protect systems and data from threats.

Legal Requirements

- Implement appropriate security controls.
- Report data breaches promptly to authorities.
- Avoid negligent handling of sensitive information.

Legal Liability

Negligence in cybersecurity—such as poor encryption, inadequate firewalls, or ignored vulnerabilities—can make professionals legally accountable for damages or losses.

F. Electronic Evidence and Digital Forensics

As cybercrime increases, courts now rely on digital evidence in legal proceedings.

Legal Considerations

- Admissibility of digital evidence in court.
- Proper collection, preservation, and chain of custody.
- Ethical limits of surveillance and digital investigation.

IT professionals working in forensic roles must respect privacy rights and due process during investigations.

G. Freedom of Information and Censorship

Information freedom laws promote public access to data held by governments, while censorship laws restrict harmful or unlawful content.

Examples

- **Freedom of Information (FOI) Act (Nigeria):** Grants citizens the right to access public records.
- **Cybercrime Act Sections 23–26:** Restrict hate speech, defamation, and child pornography.

Ethical Balance

IT professionals must balance freedom of expression with legal restrictions on harmful or false digital content.

H. Employment and Workplace Laws in IT

Legal issues also arise in the workplace environment where computing professionals operate.

Key Issues

- Employee monitoring and privacy rights.
- Workplace harassment through digital platforms.

- Whistleblower protection for reporting unethical IT practices.
- Equal opportunity and non-discrimination in technology workplaces.

Legal Frameworks

- Labour Acts and Employment Regulations.
- Anti-Discrimination and Harassment Laws.
- Whistleblower Protection Policies.

International Legal Perspectives

With globalisation, computing activities often transcend national borders.
This introduces cross-border legal complexities, such as:

- **Jurisdictional conflicts** in cybercrime investigations.
- **Data transfer regulations** between countries.
- **Global software licensing enforcement.**
- **International cooperation** through treaties like the Budapest Convention.

IT professionals must be aware of international standards to ensure compliance in global operations.

Ethical Principles Underpinning Legal Compliance

Legal compliance is closely tied to ethical responsibility in IT. Ethical codes from bodies such as the IEEE, ACM, and NCS emphasize:

Ethical Principle	Legal Relevance
Integrity	Honesty in documentation, contracts, and code development.
Confidentiality	Compliance with data protection and privacy laws.
Accountability	Taking responsibility for the legal consequences of actions.
Respect for Property	Observing intellectual property and licensing rights.
Professionalism	Avoiding negligence and adhering to regulatory standards.

Ethics and law together ensure that IT practice contributes positively to society.

Challenges in Enforcing IT Laws

Despite the existence of many laws, enforcement remains difficult due to:

- Rapid technological advancement outpacing legislation.
- Jurisdictional barriers in global cyberspace.
- Limited awareness among IT practitioners and users.
- Weak institutional enforcement and inadequate digital evidence handling.

Thus, ethical self-regulation and continuous professional education are essential complements to legal enforcement.

Best Practices for IT Professionals

To operate ethically and legally, IT professionals should:

1. Stay informed about current IT laws and regulations.
2. Ensure data protection and security compliance in all projects.
3. Respect software licensing agreements.

4. Maintain clear documentation for transparency and accountability.
5. Report cyber incidents and unethical conduct responsibly.
6. Participate in professional training and continuous learning on IT law and ethics.

ORGANISATIONAL CONTEXT

The organisational context in computing refers to the environment, structure, culture, policies, and practices of organisations that influence the ethical and legal responsibilities of IT professionals. Organisations provide the framework within which technology is designed, deployed, and managed. Understanding this context is essential for IT professionals to ensure ethical decision-making, legal compliance, and professional accountability.

In IT, the organisational context determines:

- How ethical policies are implemented.
- How legal requirements are interpreted and enforced.
- How IT professionals interact with stakeholders and manage risks.

Components of Organisational Context

The organisational context can be understood through the following key components:

a. Organisational Structure

- Defines **hierarchy, roles, and responsibilities**.
- Influences decision-making authority, including who is responsible for ethical or legal compliance.
- Examples: Centralised IT governance vs. decentralised IT departments.

Implication: Clear reporting lines help ensure accountability for ethical and legal decisions in IT projects.

b. Organisational Culture

- The **values, norms,** and **behaviours** promoted within an organisation.
- Culture influences ethical behaviour, compliance, and employee engagement.

- A culture of integrity encourages IT professionals to report misconduct and follow ethical standards.

Example: Companies with a strong ethical culture enforce policies on data privacy, cybersecurity, and intellectual property protection.

c. Policies and Procedures

- Formal documents that guide behaviour, such as codes of ethics, IT security policies, and data handling protocols.
- Procedures outline how tasks should be performed to comply with legal and ethical standards.

Importance: Policies provide clarity, reduce ambiguity, and protect both the organisation and IT professionals from legal liability.

d. Governance and Leadership

- Leadership sets the tone at the top for ethical and legal compliance.
- Governance frameworks (e.g., ITIL, COBIT) establish accountability mechanisms and risk management structures.
- Ethical leaders model behaviours, enforce compliance, and encourage transparency.

Example: CEOs and IT managers can mandate adherence to GDPR or local data protection laws.

e. Stakeholder Expectations

- Organisations must balance the interests of multiple stakeholders, including employees, clients, regulators, and the public.

- Stakeholder expectations often influence ethical and legal decision-making in IT, e.g., protecting user privacy or ensuring fair use of AI systems.

3. Ethical Issues in Organisational Context

Organisations often present ethical challenges in IT practice:

Ethical Issue	Description	Organizational Implication
Data Privacy	Handling sensitive user information responsibly	Policies must ensure confidentiality and consent
Conflict of Interest	Decisions influenced by personal gain	Requires transparency and governance to avoid breaches
Whistleblowing	Reporting unethical or illegal practices	Organisational support and protection are needed for whistleblowers
Resource Misuse	Unauthorised use of IT assets	Policies enforce the ethical use of organisational resources

Legal Issues in Organisational Context

Organisations are responsible for ensuring IT operations **comply with applicable laws**, such as:

1. **Data Protection and Privacy Laws** – GDPR, NDPR, HIPAA.
2. **Cybercrime Legislation** – Unauthorised access, malware, fraud.
3. **Intellectual Property Laws** – Software licensing, copyright, and patent compliance.
4. **Employment and Labor Laws** – Monitoring, workplace privacy, and equal opportunity.

5. **Contractual Obligations** – Adhering to service agreements and client contracts.

Organisational Relevance: Legal compliance is enforced through internal audits, IT governance frameworks, and risk management systems.

Risk Management in Organisational Context

Organisations face risks related to IT ethics and law, including:

- **Operational Risks:** System failures, downtime, and security breaches.
- **Reputational Risks:** Public backlash due to ethical lapses or data leaks.
- **Legal Risks:** Fines, penalties, and lawsuits for non-compliance.

Risk Mitigation Strategies:

- Establishing IT governance frameworks (e.g., ISO 27001, COBIT).
- Implementing security and compliance audits.
- Training employees on ethical and legal responsibilities.
- Creating incident response and reporting procedures.

Role of IT Professionals in Organisational Context

IT professionals must operate within the organisational context by:

1. **Adhering to Policies:** Follow organisational guidelines and legal requirements.
2. **Promoting Ethical Behaviour:** Advocate for fairness, transparency, and accountability.
3. **Managing Risks:** Identify, assess, and mitigate ethical and legal risks in IT operations.

4. **Supporting Governance:** Participate in IT governance and compliance initiatives.
5. **Communicating Responsibly:** Ensure clarity and transparency in reports, documentation, and stakeholder interactions.

Example: An IT security officer ensures that cybersecurity policies are implemented while also respecting employee privacy rights.

Challenges in Organisational Context

- **Conflicting Policies:** Internal policies may conflict with external legal requirements.
- **Cultural Resistance:** Employees may resist ethical or compliance policies.
- **Resource Constraints:** Budget or staffing limitations can hinder ethical IT implementation.
- **Rapid Technological Change:** New technologies may create gaps in organisational policies.

Solution: Continuous training, clear communication, and adaptive policies are critical.

Best Practices for Ethical and Legal IT Practice in Organisations

1. Develop clear codes of conduct and IT policies aligned with legal standards.
2. Encourage leadership to model ethical behaviour.
3. Provide training programs for staff on ethics, law, and compliance.
4. Implement IT governance frameworks for accountability and risk management.

5. Foster a culture of transparency and reporting to handle ethical dilemmas.
6. Continuously review and update policies to address emerging technologies and legal changes.

In summary, the organizational context shapes the ethical and legal responsibilities of IT professionals. Understanding organizational structures, culture, governance, policies, and stakeholder expectations ensures that technology is deployed responsibly, lawfully, and ethically. By navigating this context effectively, IT professionals contribute to organizational integrity, compliance, and societal trust in digital systems.

PROFESSIONAL AND ETHICAL ISSUES AND RESPONSIBILITIES – RELATIONSHIPS WITH PROFESSIONAL SOCIETIES

Professional societies play a critical role in shaping the ethical, legal, and professional standards of Information Technology (IT) practice. For IT professionals, relationships with professional societies guide them on:

- Ethical behaviour in professional practice.
- Legal compliance and awareness.
- Continuing professional development (CPD).
- Networking, advocacy, and global standards alignment.

Adhering to professional codes and engaging with societies ensures accountability, credibility, and integrity in IT practice.

Meaning of Professional Societies

Professional societies are organisations composed of individuals in a specific profession who seek to:

- Promote excellence and best practices.
- Advance knowledge and research in the field.
- Establish and enforce ethical codes and professional standards.
- Provide certification, training, and professional recognition.

Examples in IT:

- IEEE (Institute of Electrical and Electronics Engineers)
- ACM (Association for Computing Machinery)
- ISACA (Information Systems Audit and Control Association)
- NCS (Nigeria Computer Society)

Professional Responsibilities of IT Practitioners

IT **professionals are accountable to their profession, employers, clients, and society. Key responsibilities include:**

1. **Integrity and Honesty:** Presenting accurate information and acknowledging mistakes.
2. **Competence:** Maintaining and updating technical knowledge.
3. **Confidentiality:** Protecting sensitive client, user, and organisational information.
4. **Accountability:** Taking responsibility for the social, ethical, and legal impacts of technology.
5. **Fairness and Respect:** Avoiding discrimination and promoting inclusivity in professional practice.
6. **Professional Development:** Actively participating in training, seminars, and certifications offered by professional societies.

Ethical Issues in Relationships with Professional Societies

Professional societies provide codes of ethics and standards of practice. Ethical issues arise when IT professionals fail to adhere to these guidelines:

Ethical Issue	**Example**	**Professional Society Role**
Conflict of Interest	Taking personal advantage from a client contract	Codes of conduct discourage and regulate conflicts
Misrepresentation of Skills	Claiming certifications or skills not possessed	Societies provide verification and professional accreditation

Plagiarism or IP Theft	Using software, algorithms, or research without attribution	Societies enforce intellectual property respect and ethical research
Non-Compliance with Codes	Ignoring ethical codes in practice	Professional societies may sanction or revoke membership

Benefits of Professional Society Membership

Membership in professional societies offers significant ethical, professional, and practical advantages:

1. **Ethical Guidance:** Access to codes of conduct, ethical guidelines, and case studies.
2. **Legal Awareness:** Information on laws, regulations, and compliance frameworks relevant to IT.
3. **Networking:** Opportunities to collaborate with global IT professionals.
4. **Professional Recognition:** Certification and accreditation enhance career credibility.
5. **Continuous Learning:** Access to workshops, seminars, journals, and technical publications.
6. **Advocacy and Influence:** Participation in policy development and standard-setting processes.

Professional and Ethical Responsibilities Towards Societies

IT professionals have obligations to their societies, including:

1. **Upholding the Reputation of the Profession:** Avoiding actions that harm the profession's credibility.
2. **Compliance with Society Codes:** Adhering to published codes of ethics and practice.

3. **Participation:** Contributing expertise to society committees, events, and standard-setting initiatives.
4. **Mentorship:** Guiding junior professionals and students ethically and responsibly.
5. **Reporting Misconduct:** Informing societies about unethical practices by members or organisations.

Challenges in Professional Society Engagement

While professional societies provide guidance, challenges may arise:

- **Resource Constraints:** Membership fees or access to resources may be prohibitive.
- **Jurisdictional Limitations:** National societies may not enforce standards globally.
- **Ethical Conflicts:** Societal codes may differ from organisational policies or local laws.
- **Compliance Fatigue:** Professionals may ignore codes due to workload or perceived irrelevance.

Mitigation: Active engagement, continuous professional development, and ethical reflection are critical.

Examples of Codes of Ethics from Professional Societies

IEEE Code of Ethics

- Accept responsibility for professional actions.
- Avoid conflicts of interest.
- Ensure honesty, fairness, and respect.
- Improve understanding of technology's impact on society.

ACM Code of Ethics

- Contribute to society and human well-being.

- Avoid harm to others.
- Be honest and trustworthy.
- Respect privacy and confidentiality.
- Honour intellectual property rights.

ISACA Code of Professional Ethics

- Uphold the integrity, objectivity, and competence in audit, control, and security roles.
- Ensure responsible stewardship of information systems.
- Protect confidential and sensitive information.

Integrating Society Guidance into Professional Practice

IT professionals should integrate professional society guidance into practice by:

1. **Regularly Reviewing Codes of Ethics:** Staying updated with amendments.
2. **Aligning Organizational Policies:** Ensuring company policies reflect society standards.
3. **Applying Ethical Decision-Making:** Using society guidelines to resolve dilemmas.
4. **Reporting and Accountability:** Documenting adherence and reporting breaches.
5. **Professional Development:** Leveraging society resources for technical, ethical, and legal competence.

In summary, the relationship between IT professionals and professional societies is central to ethical and legal responsibility in computing.

- Societies provide standards, guidance, and support that help professionals navigate ethical dilemmas and legal obligations.

- Membership encourages continuous learning, accountability, and professional integrity.
- Ethical adherence, participation, and advocacy strengthen both the profession and societal trust in technology.

Ultimately, active engagement with professional societies ensures that IT practice aligns with global ethical and legal expectations, promoting both individual and collective professional excellence.

HOW DO PROFESSIONAL BODIES DECIDE WHAT ACTS ARE APPROPRIATE AND INAPPROPRIATE?

Your profession decides what actions are appropriate and inappropriate through three things. They are:

- ✓ Codes of conduct, codes of practice and standards.
- ✓ Computer law.
- ✓ Ethical decision making.

1. Codes of conduct, codes of practice and standards

Professional bodies will always create codes of conduct and codes of practice. The Computer Professionals Registration Council of Nigeria (CPN) code of conduct outlines what is expected as a level of professional conduct, professional integrity, that you, as a professional, pay due heed to the public interest, that you are faithful to your employer and professional field, and it also governs your technical competence and impartiality.

A code of good practice refers to how competent you are. The British Computer Society code of practice insists that you pay due attention to the personal requirements of not just yourself or your clients, but also those people who might be working for you. The code of practice outlines that you pay due heed to the organisation and management issues involved in your work, how you should undertake contract work and pay attention to privacy, security and integrity. The code of good practice covers system development, system implementation and elements of live systems.

International standards can also govern your professional activities. There are many international standards on quality management and quality assurance, the functional safety of electronic systems, and the security of information management systems. Many more standards exist.

2. Computer law

In most countries there is a considerable body of law that can apply to computer Professionals. They are:

- ✓ Contract law.
- ✓ Intellectual property law.
- ✓ Data protection law.
- ✓ Computer misuse law.
- ✓ Computer evidence.

Contract Law

If you intend to set yourself up in business you should become familiar with contract law. The most important thing to point out is that the ownership of intellectual property of something you developed depends very much on your role when you developed it. The key phrase is "I did it in the course of my employment" – if you develop something in the course of your employment the ownership generally belongs to your employer. If you do something as a contractor, invariably the ownership belongs to you – you have not been employed; you have been brought in and contracted to develop something.

If you are producing computer software with the intent of licensing the software to whoever commissioned it then you own the software. If you produce a piece of bespoke software then, unless you explicitly state in the contract that you intend to license the software, whoever commissioned it has paid you for, and therefore owns, what you have developed.

Contractual duties which you cannot avoid are:

- Fidelity – dealing honestly and faithfully with people.
- Confidence – how you respect the confidences that are given to you.
- Culpability – the fact that you could be taken to court if you are negligent.
- You CANNOT contract out of reasonable liabilities.

Intellectual Property Law

Intellectual property law centres on two main moral rights. The right of paternity, the right to be recognised as the person who created something. And the right of integrity, the right not to have your work tampered with by someone else. Copyright protects original works, sound recordings, and typographical layouts. Copyright lasts for 70 years after the death of the person who created it. In the case of a computer-generated work, where the author did not die, copyright expires 50 years after the work's creation. Patents protect work for 20 years. Patented ideas are generally novel and not obvious; the important thing to remember is that if you patent something, you are expected to use it. In the UK, a design right for designs such as printed circuit board layouts is recognised.

Data Protection Law

There are many statutes of data protection law; because of this it is difficult to distil general principles from these many statutes. The most important thing to remember is that the subject of personal data has the right to view and correct that data. Personal data should be accurate, adequate – sufficient for its purpose, relevant, and kept up to date. Personal data should not be kept for any longer than is necessary. All appropriate technical and organizational measures should be taken against unauthorized or unlawful processing of personal data and against accidental loss or destruction of personal data. For computer data you should make sure that:

· You have appropriate backups of the data.

· You take appropriate security measures to secure the data.

Computer Misuse Law

Computer misuse law refers to unauthorised access and unauthorised modification. Unauthorised access of systems, programs, and data is unacceptable while unauthorised modifications through editing and deleting are forbidden. It has been recognised that, through the use of computers, it is possible to commit a crime in a country without actually being present in that country. Modern legislation can now state that it is irrelevant if you are in the country where the crime took place as long as it can be proved that you were responsible for the crime. 10

Computer Evidence

Computer evidence concerns what evidence is permissible in a court of law. Evidence may be invalidated if it is viewed after the event. An example of this is if you view log files with an editor after an intrusion, this will invalidate the logs as evidence as it is deemed that they might have been altered after the intrusion. Under normal circumstances specialists are brought in to follow audit trails back to the place of the attack. Amateurs could invalidate evidence or unwittingly tip off perpetrators.

3. Ethical Decision Making

In discussing ethical decision making, the areas which we will cover are:

- ✓ It is not just about deciding what is right and what is wrong.
- ✓ Moral systems and principles.
- ✓ Stakeholder analysis.
- ✓ Six useful tests.

Ethics is not just about what is right and what is wrong. We can become absorbed in the technicalities to the point that it becomes so complicated that we cannot explain things in simple terms. When things go wrong and you are responsible in some way the public may seek assurance from you that the technical decisions which were taken paid due heed to the interests of the public. It is crucial that

technologists can provide clear and understandable justifications for their motives, decisions, and actions.

A way of convincing people that you have their best interests in mind is by citing moral systems. Citing moral systems and principles can help to provide assurance that one's motives, etc. were of good intent. It is important to remember that even though these moral systems and principles have sound foundations, not everybody will be reassured by such references.

A **stakeholder analysis approach** is a more organised approach. Stakeholder analysis can help you to arrive at sound decisions and provide justifications. Any person, group or organisation that could be affected by the decision is a stakeholder. Stakeholders are not always easy to identify, some are only affected very indirectly. Stakeholder analysis tabulates the alternative decisions and the stakeholders, noting the effect of each alternative on each stakeholder e.g. very good, good, neutral, bad, or very bad.

Six useful tests which you can apply if you are presented with some kinds of dilemma are:

1. The Golden Rule – how would you like it if what you are proposing to do was done to you? **Mt 7: 12** *(Therefore all things whatsoever ye would that men should do to you, do ye even so to them: for this is the law and the prophets.*)

2. The legality test – Is what you are proposing to do legal?

3. The smell test – do you have that feeling in your stomach that it's right or that it's wrong?

4. The parent test – would you tell your parents that you are planning to do this?

5. The media test – how would you defend this on television?

6. The market test – is the solution that you have come up with so good that you should think about selling it?

ETHICS

Every society forms a set of rules that establishes the boundaries of generally accepted behaviour. These rules are often expressed in statements about how people should behave, and the individual rules fit together to form the moral code by which a society lives.

A moral code is a set of rules that establishes the boundaries of generally accepted behaviour. The term **morality** refers to social conventions about right and wrong that are so widely shared that they become the basis for an established consensus.

Ethics is a set of beliefs about right and wrong behaviour within a society. Ethical behaviour conforms to generally accepted norms—many of which are almost universal. Behaviours that are considered ethical in one culture may be unethical in another. For example, software piracy —a form of copyright infringement that involves making copies of software or enabling others to access software to which they are not entitled- may be acceptable in some cultures and yet considered very unethical in others.

People generally develop habits that make it easier for them to choose between what society considers good or bad. A **virtue** is a habit that inclines people to do what is acceptable, for example, fairness, kindness, and honesty. A vice is a habit of unacceptable behaviour, such as greed, lying, and cheating.

The Importance of Integrity

A person who acts with integrity acts in accordance with a personal code of principles. One approach to acting with integrity—one of the cornerstones of ethical behaviour—is to extend to all people the same respect and consideration that one expects to receive from others.

If people are consistent and act with integrity, they apply the same moral standards in all situations. However, there are situations in which it might be difficult to be consistent in applying moral standards because many ethical dilemmas are not as simple as right versus wrong but involve choices between right versus right.

The Difference between Morals, Ethics, and Laws

Morals refer to one's personal beliefs about right and wrong, while the term ethics describes standards or codes of behaviour expected of an individual by a group (nation, organisation, profession) to which an individual belongs.

Law is a system of rules that tells what can and cannot be done. Laws are enforced by a set of institutions (the police, courts, law-making bodies), and legal acts are acts that conform to the law.

Moral acts conform to what an individual believes to be the right thing to do.

Ethics in Information Technology

This is the ability to capture and store vast amounts of personal data, and greater reliance on information systems in all aspects of life has increased the risk that information technology will be used unethically. Some examples of Information technology ethics challenges include:

a. Employers monitoring email and Internet usage by employees.

b. Illegal downloading of music and movies.

c. Hackers breaking into financial and retail institutions, such as the recent data breaches reported by several organisations across the world.

d. Use of big data to track people's online purchases and activities.

WHISTLE-BLOWING

Whistle-blowing refers to the act of an employee or professional exposing unethical, illegal, or harmful activities within an organisation. In the field of Information Technology (IT), it involves reporting issues such as data breaches, privacy violations, software piracy, misuse of organisational resources, or cybercrime by insiders.

Ethical Basis of Whistle Blowing

Whistle-blowing is grounded in ethical responsibility — the duty to protect the public, uphold justice, and maintain professional integrity. An IT professional must balance loyalty to the organisation with moral duty to society when wrongdoing threatens the public interest.

Conditions for Ethical Whistle Blowing

Whistle-blowing is considered ethical when:

1. The wrongdoing causes serious harm (e.g., data theft, security breach).
2. The employee has credible evidence of misconduct.
3. Internal reporting channels have been exhausted.
4. Disclosure is made in good faith and not for personal gain.
5. The whistleblower is prepared to face professional or legal consequences.

Legal Protection for Whistleblowers

Many jurisdictions have laws protecting whistleblowers:

- **Nigeria's Whistle Blower Protection Policy (2016):** Encourages reporting of financial or ethical misconduct.

- **GDPR / NDPR:** Protects individuals who disclose data misuse or privacy violations.
- **Corporate policies:** Many organizations include whistle blowing mechanisms in their **IT governance frameworks**.

Whistle Blowing in IT Practice

Common IT-related whistle blowing cases include:

- Unauthorized data collection or surveillance.
- Hiding cybersecurity vulnerabilities from regulators.
- Manipulation of software or algorithmic bias.
- Misuse of public or organizational IT funds.

Ethical IT professionals must act transparently, responsibly, and legally when exposing such misconduct.

Ethical Dilemma

Whistle blowing often creates tension between:

- Loyalty to employer and
- Obligation to public interest.

Professionals must evaluate the magnitude of harm, available reporting channels, and legal safeguards before taking action.

In summary, the context of professional ethics and legal issues in IT, whistleblowing represents the moral courage to uphold truth, accountability, and justice in digital environments.
An ethical whistleblower helps protect data integrity, public trust, and professional credibility, ensuring that technology serves society responsibly.

WORKPLACE ISSUES (HARASSMENT, DISCRIMINATION)

In the IT workplace, harassment and discrimination undermine professionalism, productivity, and ethical integrity. They occur when individuals are treated unfairly or subjected to hostile behavior based on gender, race, religion, disability, age, or other personal attributes.

Harassment in IT

Harassment includes unwelcome behavior—verbal, physical, or digital—that creates an intimidating or offensive work environment. Examples:

- Cyberbullying or offensive emails.
- Sexual or gender-based remarks.
- Abuse of power in supervision or promotions.

Ethical principle: Every IT professional has the right to dignity, respect, and safety at work.

Discrimination in IT

Discrimination involves unfair treatment or exclusion in hiring, training, promotion, or pay based on non-merit factors. Examples:

- Gender bias in tech roles.
- Denying opportunities due to disability or ethnicity.
- Unequal access to professional development.

Legal frameworks: Labour laws, Equal Opportunity Acts, and workplace diversity policies prohibit discrimination.

Professional and Ethical Response

- Promote inclusivity and diversity in IT teams.

- Report harassment through official channels.
- Uphold organizational codes of conduct and professional ethics (ACM, IEEE).
- Create a respectful, collaborative, and supportive work culture.

In summary, harassment and discrimination violate both ethical standards and legal principles in IT.
An ethical IT professional must respect human dignity, advocate fairness, and help build workplaces that reflect equality and professionalism.

IDENTIFY THEFT

Identity theft is the unauthorized acquisition and use of another person's personal information—such as name, password, credit card, or biometric data—to commit fraud or other crimes. In IT, it represents both a legal violation and an ethical breach of trust and privacy.

Ethical and Legal Dimensions

- **Ethical Issue:** Misuse of identity data violates principles of honesty, confidentiality, and respect for persons.
- **Legal Issue:** Most nations, including Nigeria (under the **Cybercrimes Act, 2015**), criminalize identity theft and related online fraud.
- **Professional Duty:** IT professionals must implement strong data protection and privacy measures to prevent identity misuse.

Common Forms of Identity Theft

- **Phishing and Social Engineering** – tricking users to reveal credentials.
- **Database Breaches** – stealing stored personal data.
- **Financial Fraud** – using stolen identities for transactions.
- **Impersonation on Social Media** – damaging reputation or spreading misinformation.

Ethical Responsibilities of IT Professionals

- Safeguard user data with encryption and access control.
- Educate users on safe digital practices.
- Report and mitigate breaches promptly.
- Adhere to data privacy laws (e.g., **NDPR**, **GDPR**).

In summary, identity theft undermines trust, privacy, and security in digital systems. Ethical IT professionals must act as custodians of

personal data, ensuring its responsible use, legal compliance, and protection from abuse.

ETHICAL HACKING

Ethical hacking, also known as penetration testing or white-hat hacking, is the practice of legally probing computer systems, networks, and applications to identify vulnerabilities before malicious actors exploit them. Unlike malicious hackers (black-hat hackers), ethical hackers operate within the law, follow organisational policies, and adhere to professional codes of conduct.

In the context of professional ethics and legal issues, ethical hacking illustrates the balance between technical expertise, ethical responsibility, and legal compliance in IT practice. Ethical hacking is a critical intersection of technical skill, ethics, and law in the IT profession.

- It allows organisations to pre-empt cyber threats while respecting legal boundaries.
- IT professionals engaged in ethical hacking must obtain consent, act responsibly, and maintain confidentiality.
- Ethical hacking contributes to trust, security, and professional integrity, reinforcing the societal role of IT professionals.

Ultimately, ethical hacking demonstrates that security and ethics are inseparable in modern IT practice.

Purpose of Ethical Hacking

Ethical hacking serves several key purposes:

1. **Security Assessment:** Identify weaknesses in systems, applications, and networks.
2. **Risk Mitigation:** Reduce potential damage from cyberattacks.
3. **Compliance:** Ensure adherence to legal and regulatory requirements (e.g., GDPR, NDPR).
4. **Training and Awareness:** Educate employees about security best practices.

5. **Incident Prevention:** Proactively strengthen defenses before breaches occur.

Types of Hackers

Hacker Type	Description	Legal/Ethical Status
White-Hat Hacker	Performs ethical hacking with authorization	Legal, ethical
Black-Hat Hacker	Exploits systems for personal or financial gain	Illegal, unethical
Gray-Hat Hacker	Exploits vulnerabilities without permission but may report them	Legally ambiguous, ethically debated
Red Team / Blue Team	Simulated attack (Red) and defense (Blue) exercises	Legal when authorized, used for security training

Ethical Hacking Methodologies

Ethical hackers follow structured processes to ensure legality and effectiveness:

1. **Reconnaissance:** Gathering information about the target system.
2. **Scanning and Enumeration:** Identifying open ports, services, and potential vulnerabilities.
3. **Gaining Access:** Exploiting vulnerabilities in a controlled and legal manner.
4. **Maintaining Access:** Testing how an attacker could persist, without causing harm.
5. **Analysis and Reporting:** Documenting findings and recommending remediation.

Ethical Requirement: Every step must be authorised, documented, and limited to the agreed scope.

Legal Frameworks Governing Ethical Hacking

Ethical hacking must comply with laws and organisational policies:

1. **Computer Misuse Laws:** Unauthorised access or testing without consent is illegal.
2. **Cybercrime Legislation:** (e.g., Nigeria Cybercrime Act 2015) prohibits unauthorised intrusion.
3. **Data Protection Laws:** (e.g., GDPR, NDPR) ensure personal data is not exposed during testing.
4. **Contractual Agreements:** Penetration testing should be backed by formal authorisation agreements and non-disclosure agreements (NDAs).

Key Principle: Ethical hacking is legal only when permission is explicitly granted by the system owner.

Ethical Principles in Hacking

Ethical hacking aligns with professional and societal values:

Principle	Application in Ethical Hacking
Consent	Always obtain explicit permission from the system owner.
Integrity	Report all findings honestly without manipulation.
Confidentiality	Protect sensitive data discovered during testing.
Accountability	Accept responsibility for the outcomes of testing activities.
Non-Maleficence	Avoid causing harm to systems, data, or users.

Professional Responsibilities of Ethical Hackers

Ethical hackers must maintain professionalism and integrity:

1. **Authorization:** Only hack systems with documented approval.
2. **Scope Management:** Follow defined boundaries to prevent unintended damage.
3. **Reporting:** Clearly communicate vulnerabilities, risks, and recommended remedies.
4. **Continuous Learning:** Stay updated on emerging threats, vulnerabilities, and security tools.
5. **Collaboration:** Work with IT teams to strengthen organizational security.

Benefits of Ethical Hacking

- **Enhanced Security:** Identifies and mitigates vulnerabilities proactively.
- **Legal Compliance:** Helps organizations comply with regulatory requirements.
- **Trust Building:** Demonstrates responsible cybersecurity practices to stakeholders.
- **Cost Reduction:** Prevents financial and reputational losses due to breaches.
- **Skill Development:** Strengthens IT professionals' expertise in security and risk management.

Risks and Challenges in Ethical Hacking

Risk/Challenge	Description	Mitigation
Unauthorized Access	Hacking outside the agreed scope	Clear contracts and scope definition
Data Exposure	Accidental leakage of sensitive data	Secure handling and encryption of test data
System Damage	Testing may disrupt operations	Testing in controlled environments or using backups
Legal Liability	Non-compliance with laws can lead to prosecution	Adherence to cyber laws and written authorizations
Ethical Dilemmas	Reporting vs. exploiting vulnerabilities	Following professional codes of ethics

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Ethical Hacking in Organisational Context

Organisations often employ ethical hacking as part of information security management:

- Conducting penetration tests regularly.
- Performing vulnerability assessments before deploying new systems.
- Training staff on security awareness using simulated attacks.

- Integrating ethical hacking findings into risk management frameworks.

Ethical hacking aligns with organisational ethics, legal compliance, and corporate governance.

Professional Societies and Ethical Hacking

Professional societies provide guidelines, certifications, and standards:

- **EC-Council Certified Ethical Hacker (CEH):** Provides structured training on legal and ethical hacking.
- **ISACA:** Offers guidance on IT audit, cybersecurity, and governance.
- **IEEE / ACM Codes of Ethics:** Emphasise responsible, lawful, and ethical IT practice.

Membership in these societies encourages ethical behaviour and legal compliance for penetration testers.

PRIVACY AND CIVIL LIBERTIES

Privacy refers to the right of individuals to control how their personal information is collected, stored, and used. Civil liberties are fundamental rights—such as freedom of expression, association, and protection from surveillance—that ensure human dignity in a digital society. In IT, respecting privacy and civil liberties is an essential ethical and legal responsibility.

Ethical and Legal Perspectives

- **Ethical View:** IT professionals must respect user confidentiality and data ownership.
- **Legal View:** Laws like the Nigeria Data Protection Regulation (NDPR) and GDPR protect users from data misuse and unauthorised surveillance.
- Breach of privacy or violation of civil liberties leads to loss of trust and legal penalties.

Threats to Privacy and Civil Liberties

- Mass data collection and surveillance

- Unauthorised data sharing
- Social media profiling
- Invasive algorithms and AI tracking

IT systems must be designed to protect individual freedoms while ensuring security and transparency.

Professional Responsibilities

- Collect only necessary data (data minimisation).
- Obtain informed consent before data use.
- Ensure transparency, accountability, and fairness in data handling.
- Advocate for ethical technology policies that balance innovation with human rights.

In summary, protecting privacy and civil liberties is central to ethical IT practice. IT professionals must uphold trust, human dignity, and digital freedom while ensuring compliance with data protection laws and professional codes of conduct.